

1. General Description

The EM74HC595A is an 8-bit serial-in/serial or parallel-out shift register with a storage register and 3-state outputs. Both the shift and storage register have separate clocks. The device features a serial input (DS) and a serial output (Q7S) to enable cascading and an asynchronous reset $\overline{MR}$ input. A LOW on $\overline{MR}$ will reset the shift register. Data is shifted on the LOW-to-HIGH transitions of the SHCP input. The data in the shift register is transferred to the storage register on a LOW-to-HIGH transition of the STCP input. If both clocks are connected together, the shift register will always be one clock pulse ahead of the storage register. Data in the storage register appears at the output whenever the output enable input ($\overline{OE}$) is LOW. A HIGH on $\overline{OE}$ causes the outputs to assume a high-impedance OFF-state. Operation of the $\overline{OE}$ input does not affect the state of the registers. Inputs include clamp diodes. This enables the use of current limiting resistors to interface inputs to voltages in excess of V_{CC} .

2. Features and Benefits

- Wide operating voltage 2.0 V to 6.0 V
- High noise immunity
- CMOS low power dissipation
- 8-bit serial input
- 8-bit serial or parallel output
- Storage register with 3-state outputs
- Shift register with direct clear
- Latch-up performance exceeds 250 mA
- Complies with JEDEC standard:
 - JESD8C(2.7 V to 3.6 V)
 - JESD7A(2.0 V to 6.0 V)
- ESD protection:
 - HBM ANSI/ESDA/JEDEC JS-001 Class 3A exceeds 6000 V
 - CDM ANSI/ESDA/JEDEC JS-002 Class C3 exceeds 2000 V

- Multiple package options
- Specified from -40 °C to +85 °C and from -40 °C to +125 °C

3.Applications

- Serial-to-parallel data conversion
- Remote control holding register

EM74HC595A

8-bit serial-in, serial or parallel-out shift register with output latches; 3-state

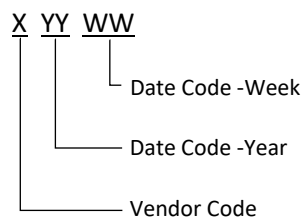
4. Ordering Information

Table 1. Ordering information

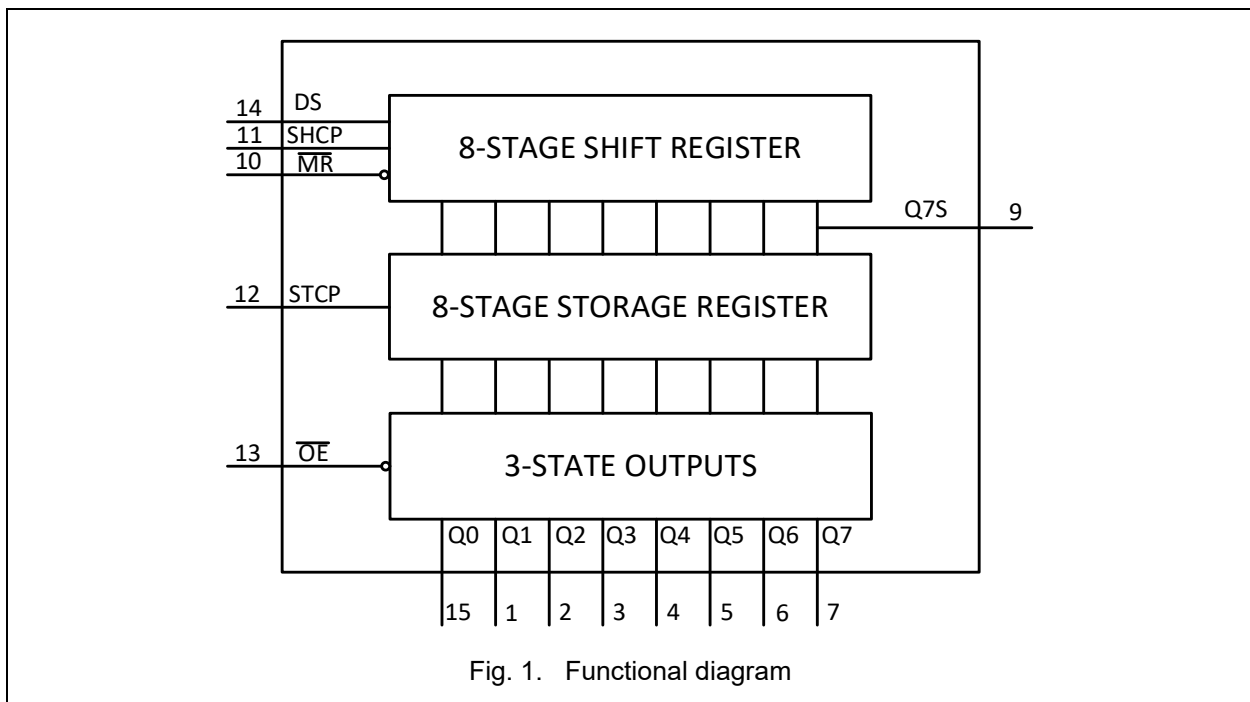
Type number	Topside marking	Package		
		Name	Description	Quantity
EM74HC595AD	HC595A YYYWW	SOP-16L	plastic small outline package; 16 leads; body width 3.9 mm	3000
EM74HC595APW	HC595A YYYWW	TSSOP-16L	plastic thin shrink small outline package; 16 leads; body width 4.4 mm	3000

MARKING INFORMATION

NOTE: YYYWW = Vendor Code and Date Code.

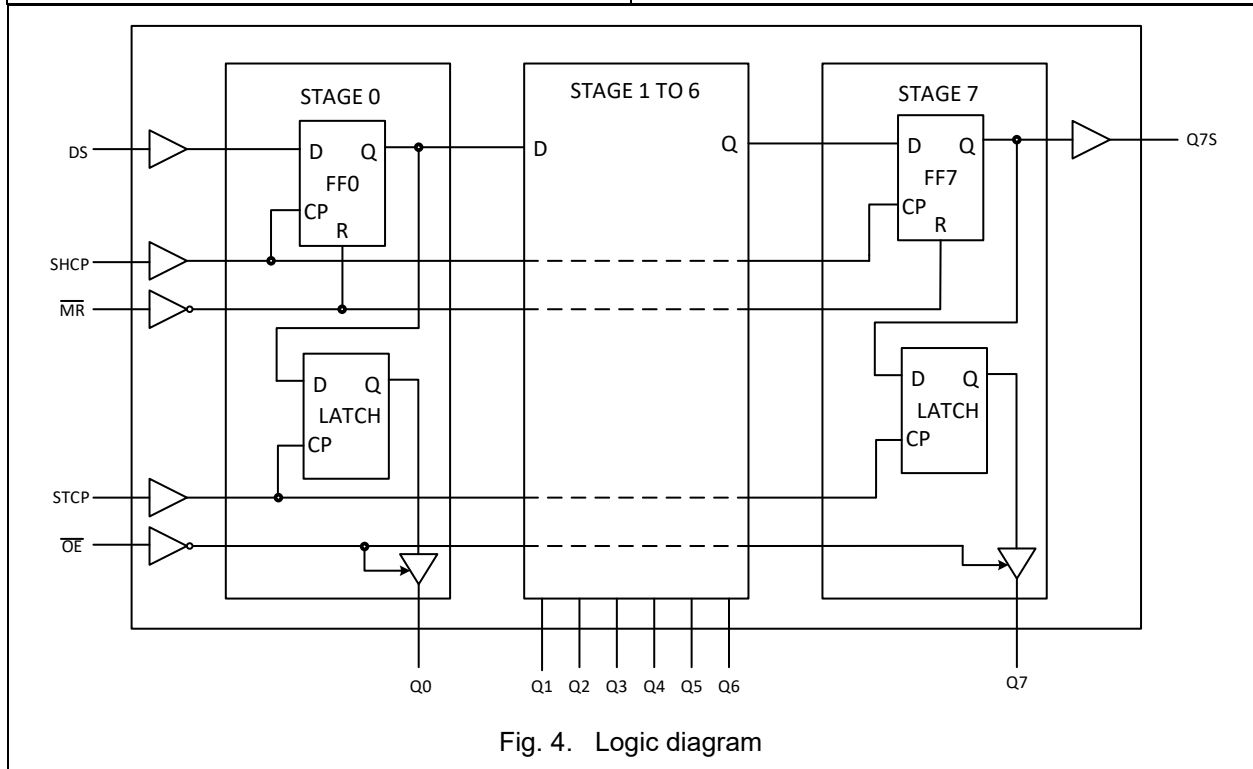
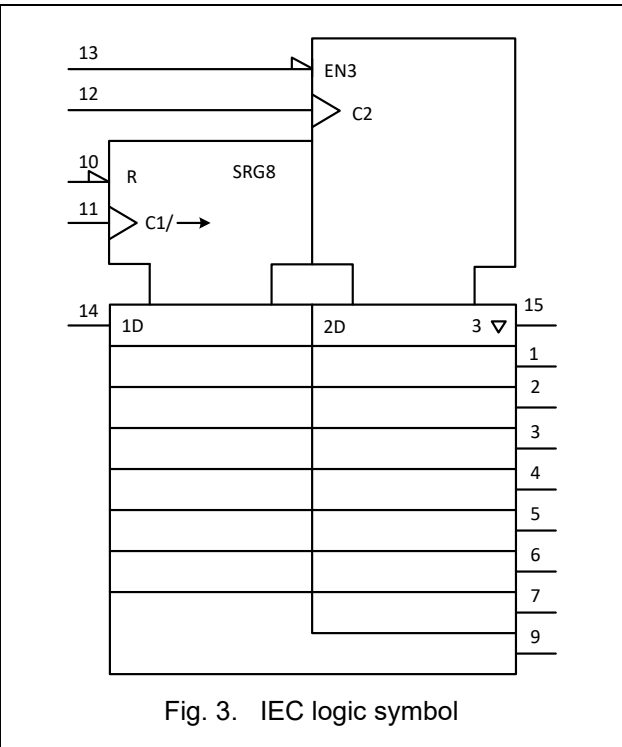
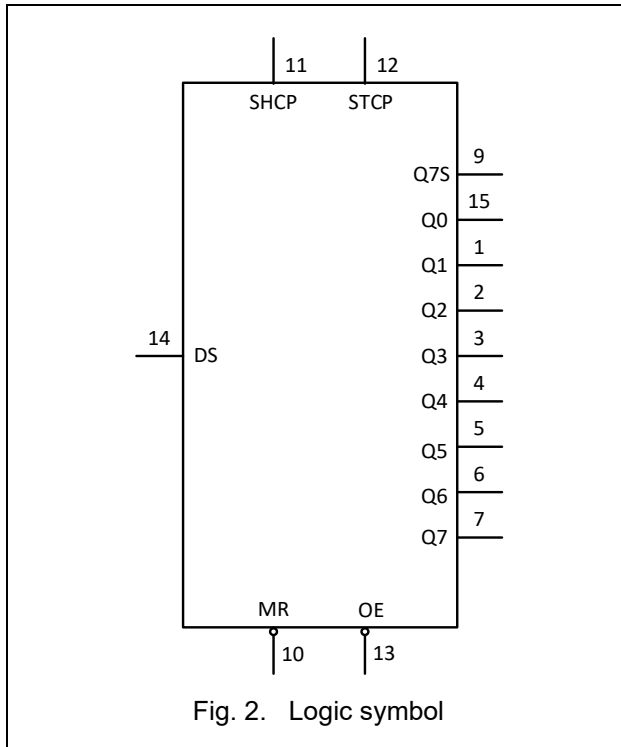


5. Function Diagram



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8-bit serial-in, serial or parallel-out shift register with output latches; 3-state

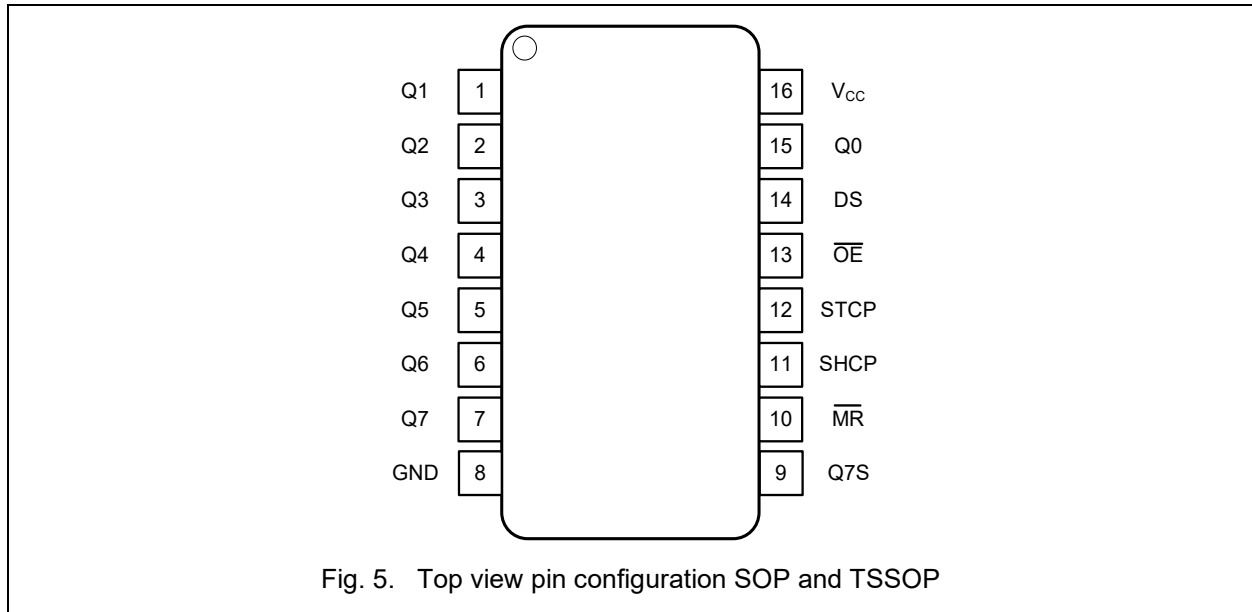


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6. Pinning Information

6.1. Pinning



6.2. Pin description

Table 2. Pin description

Symbol	Pin	Description
Q0, Q1, Q2, Q3, Q4, Q5, Q6, Q7	15, 1, 2, 3, 4, 5, 6, 7	Parallel data output
GND	8	Ground (0V)
Q7S	9	Serial data output
$\overline{\text{MR}}$	10	Master reset (active LOW)
SHCP	11	Shift register clock input
STCP	12	Storage register clock input
$\overline{\text{OE}}$	13	Output enable input (active LOW)
DS	14	Serial data input
Q0	15	Parallel data output 0
V _{CC}	16	Supply voltage

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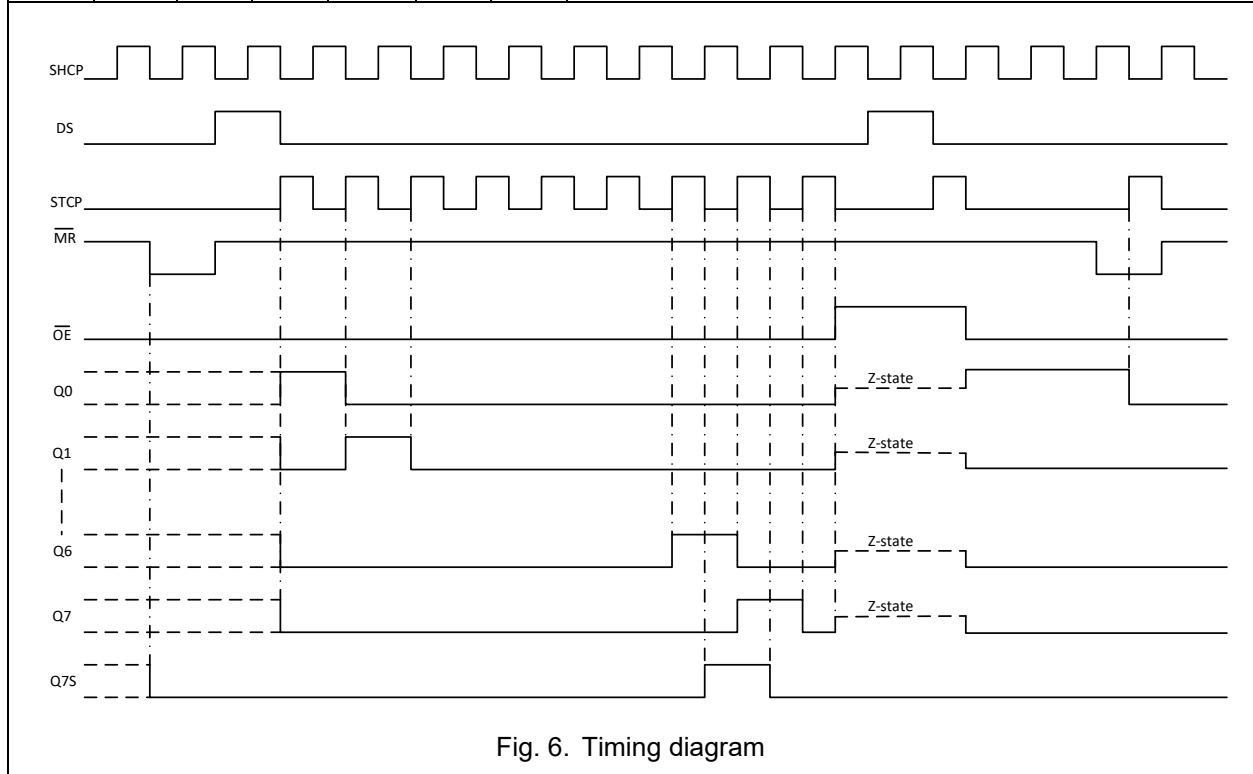
7.Functional Description

Table 3. Function table

H = HIGH voltage level; L = LOW voltage level; $\uparrow$ = LOW-to-HIGH transition;

X = don't care; NC = no change; Z = high-impedance OFF-state.

Control				Input	Output		Function
SHCP	STCP	$\overline{OE}$	$\overline{MR}$	DS	Q7S	Qn	
X	X	L	L	X	L	NC	a LOW-level on $\overline{MR}$ only affects the shift registers
X	$\uparrow$	L	L	X	L	L	empty shift register loaded into storage register
X	X	H	L	X	L	Z	shift register clear; parallel outputs in high-impedance OFF-state
$\uparrow$	X	L	H	H	Q6S	NC	logic HIGH-level shifted into shift register stage 0. Contents of all shift register stages shifted through, e.g. previous state of stage 6 (internal Q6S) appears on the serial output (Q7S).
X	$\uparrow$	L	H	X	NC	QnS	contents of shift register stages (internal QnS) are transferred to the storage register and parallel output stages
$\uparrow$	$\uparrow$	L	H	X	Q6S	QnS	contents of shift register shifted through; previous contents of the shift register is transferred to the storage register and the parallel output stages



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8. Absolute Maximum Ratings

Stresses exceeding the absolute maximum ratings may damage the device. The device may not function or be operable above the recommended operating conditions and stressing the parts to these levels is not recommended. In addition, extended exposure to stresses above the recommended operating conditions may affect device reliability. The absolute maximum ratings are stress ratings only.

Table 4. Absolute Maximum Ratings

In accordance with the Absolute Maximum Rating System (IEC 60134). Voltages are referenced to GND.

Symbol	Parameter	Conditions	Min	Max	Unit
V_{CC}	supply voltage		-0.5	7.0	V
I_{IK}	input clamping current	$V_O < -0.5 \text{ V}$ or $V_O > V_{CC} + 0.5 \text{ V}$ [1]		± 20	mA
I_{OK}	output clamping current	$V_O < -0.5 \text{ V}$ or $V_O > V_{CC} + 0.5 \text{ V}$ [1]		± 20	mA
I_O	output current	$-0.5 \text{ V} < V_O < V_{CC} + 0.5 \text{ V}$		± 35	mA
I_{CC}	supply current			70	mA
I_{GND}	ground current		-70		mA
P_{tot}	total power dissipation	$T_{amb} = -40 \text{ }^\circ\text{C}$ to $+125 \text{ }^\circ\text{C}$		500	mW
T_{stg}	storage temperature		-65	150	$^\circ\text{C}$

[1] The input and output voltage ratings may be exceeded if the input and output current ratings are observed.

9. Recommended Operating Conditions

The Recommended Operating Conditions table defines the conditions for actual device operation. Recommended operating conditions are specified to ensure optimal performance to the datasheet specifications. EnergyMath does not recommend exceeding them or designing to Absolute Maximum Ratings.

Table 5. Recommended Operating Conditions

Symbol	Parameter	Conditions	EM74HC595A			Unit
			Min	Typ	Max	
V_{CC}	supply voltage		2.0	5.0	6.0	V
V_I	input voltage		0		V_{CC}	V
V_O	output voltage		0		V_{CC}	V
T_{amb}	ambient temperature		-40	+25	+125	$^\circ\text{C}$
$\Delta t/\Delta V$	input transition rise and fall rate	$V_{CC} = 2.0 \text{ V}$			625	ns/V
		$V_{CC} = 4.5 \text{ V}$		1.67	139	ns/V
		$V_{CC} = 6.0 \text{ V}$			83	ns/V

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8-bit serial-in, serial or parallel-out shift register with output latches; 3-state

10. Static Characteristics

Table 6. Static characteristics

At recommended operating conditions. Voltages are referenced to GND (ground = 0 V).

Symbol	Parameter	Conditions	-40 °C to +85 °C			-40 °C to +125 °C		Unit
			Min	Typ[1]	Max	Min	Max	
V_{IH}	HIGH-level input voltage	$V_{CC} = 2.0 \text{ V}$	1.5	1.26		1.5		V
		$V_{CC} = 4.5 \text{ V}$	3.15	2.48		3.15		V
		$V_{CC} = 6.0 \text{ V}$	4.2	3.22		4.2		V
V_{IL}	LOW-level input voltage	$V_{CC} = 2.0 \text{ V}$		0.84	0.5		0.5	V
		$V_{CC} = 4.5 \text{ V}$		1.99	1.35		1.35	V
		$V_{CC} = 6.0 \text{ V}$		2.65	1.8		1.8	V
V_{OH}	HIGH-level output voltage	$V_I = V_{IH} \text{ or } V_{IL}$						
		all outputs						
		$I_O = -20 \mu\text{A}; V_{CC} = 2.0 \text{ V}$	1.9	2.0		1.9		V
		$I_O = -20 \mu\text{A}; V_{CC} = 4.5 \text{ V}$	4.4	4.5		4.4		V
		$I_O = -20 \mu\text{A}; V_{CC} = 6.0 \text{ V}$	5.9	6.0		5.9		V
		Q7S output						
		$I_O = -4.0 \text{ mA}; V_{CC} = 4.5 \text{ V}$	3.84	4.39		3.7		V
		$I_O = -5.2 \text{ mA}; V_{CC} = 6.0 \text{ V}$	5.34	5.88		5.2		V
		Qn bus driver outputs						
		$I_O = -6.0 \text{ mA}; V_{CC} = 4.5 \text{ V}$	3.84	4.34		3.7		V
		$I_O = -7.8 \text{ mA}; V_{CC} = 6.0 \text{ V}$	5.34	5.82		5.2		V
V_{OL}	LOW-level output voltage	$V_I = V_{IH} \text{ or } V_{IL}$						
		all outputs						
		$I_O = 20 \mu\text{A}; V_{CC} = 2.0 \text{ V}$		0	0.1		0.1	V
		$I_O = 20 \mu\text{A}; V_{CC} = 4.5 \text{ V}$		0	0.1		0.1	V
		$I_O = 20 \mu\text{A}; V_{CC} = 6.0 \text{ V}$		0	0.1		0.1	V
		Q7S output						
		$I_O = 4.0 \text{ mA}; V_{CC} = 4.5 \text{ V}$		0.08	0.33		0.4	V
		$I_O = 5.2 \text{ mA}; V_{CC} = 6.0 \text{ V}$		0.08	0.33		0.4	V
		Qn bus driver outputs						
		$I_O = 6.0 \text{ mA}; V_{CC} = 4.5 \text{ V}$		0.11	0.33		0.4	V
		$I_O = 7.8 \text{ mA}; V_{CC} = 6.0 \text{ V}$		0.13	0.33		0.4	V

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8-bit serial-in, serial or parallel-out shift register with output latches; 3-state

Symbol	Parameter	Conditions	-40 °C to +85 °C			-40 °C to +125 °C		Unit
			Min	Typ[1]	Max	Min	Max	
I_I	input leakage current	$V_I = V_{CC}$ or GND ; $V_{CC} = 6.0$ V		0.0005	± 1		± 1	μA
I_{OZ}	OFF-state output current	$V_I = V_{IH}$ or V_{IL} ; $V_{CC} = 6.0$ V ; $V_O = V_{CC}$ or GND		0.002	± 5		± 10	μA
I_{CC}	supply current	$V_I = V_{CC}$ or GND ; $I_O = 0$ A ; $V_{CC} = 6.0$ V		13.7	20		40	μA
C_i	input capacitance			1.2				pF

[1] All typical values are measured at $T_{amb} = 25^\circ C$.

11. Dynamic Characteristics

Table 7. Dynamic characteristics

Voltages are referenced to GND (ground = 0 V); for test circuit see Fig. 12.

Symbol	Parameter	Conditions	-40 °C to +85 °C			-40 °C to +125 °C		Unit
			Min	Typ[1]	Max	Min	Max	
t_{pd}	propagation delay	SHCP to Q7S; see Fig. 7 [2]						
		$V_{CC} = 2.0$ V		18.4	30		35	ns
		$V_{CC} = 4.5$ V		7.6	15		20	ns
		$V_{CC} = 6.0$ V		6.2	10		15	ns
		STCP to Q7S; see Fig. 8 [2]						
		$V_{CC} = 2.0$ V		17.8	30		35	ns
		$V_{CC} = 4.5$ V		7.4	15		20	ns
		$V_{CC} = 6.0$ V		6.1	10		15	ns
t_{PHL}	HIGH to LOW propagation delay	$\overline{MR}$ to Q7S; see Fig. 10						
		$V_{CC} = 2.0$ V		19.2	30		35	ns
		$V_{CC} = 4.5$ V		7.9	15		20	ns
		$V_{CC} = 6.0$ V		6.5	10		15	ns
t_{en}	enable time	$\overline{OE}$ to Qn; see Fig. 11 [3]						
		$V_{CC} = 2.0$ V		15.5	25		30	ns
		$V_{CC} = 4.5$ V		5.8	14		17	ns
		$V_{CC} = 6.0$ V		4.9	13		16	ns

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Symbol	Parameter	Conditions	-40 °C to +85 °C			-40 °C to +125 °C		Unit
			Min	Typ[1]	Max	Min	Max	
t_{dis}	disable time	$\overline{OE}$ to Q_n ; see Fig. 11 [4]						
		$V_{CC} = 2.0\text{ V}$		10.3	25		30	ns
		$V_{CC} = 4.5\text{ V}$		5.5	14		17	ns
		$V_{CC} = 6.0\text{ V}$		4.5	13		16	ns
t_w	pulse width	SHCP HIGH or LOW; see Fig. 7						
		$V_{CC} = 2.0\text{ V}$	25			30		ns
		$V_{CC} = 4.5\text{ V}$	19			22		ns
		$V_{CC} = 6.0\text{ V}$	16			19		ns
		STCP HIGH or LOW; see Fig. 8						
		$V_{CC} = 2.0\text{ V}$	25			30		ns
		$V_{CC} = 4.5\text{ V}$	19			22		ns
		$V_{CC} = 6.0\text{ V}$	16			19		ns
		$\overline{MR}$ LOW; see Fig.10						
		$V_{CC} = 2.0\text{ V}$	25			30		ns
		$V_{CC} = 4.5\text{ V}$	19			22		ns
		$V_{CC} = 6.0\text{ V}$	16			19		ns
t_{su}	set up time	DS to SHCP; see Fig.9						
		$V_{CC} = 2.0\text{ V}$	16			20		ns
		$V_{CC} = 4.5\text{ V}$	13			15		ns
		$V_{CC} = 6.0\text{ V}$	11			13		ns
		SHCP to STCP; see Fig.9						
		$V_{CC} = 2.0\text{ V}$	25			30		ns
		$V_{CC} = 4.5\text{ V}$	19			22		ns
		$V_{CC} = 6.0\text{ V}$	16			19		ns
t_h	hold time	DS to SHCP; see Fig.9						
		$V_{CC} = 2.0\text{ V}$	3			3		ns
		$V_{CC} = 4.5\text{ V}$	3			3		ns
		$V_{CC} = 6.0\text{ V}$	3			3		ns
t_{rec}	recovery time	$\overline{MR}$ to SHCP; see Fig. 10						
		$V_{CC} = 2.0\text{ V}$	16			20		ns
		$V_{CC} = 4.5\text{ V}$	13			15		ns
		$V_{CC} = 6.0\text{ V}$	11			13		ns

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8-bit serial-in, serial or parallel-out shift register with output latches; 3-state

Symbol	Parameter	Conditions	-40 °C to +85 °C			-40 °C to +125 °C		Unit
			Min	Typ[1]	Max	Min	Max	
$f_{\max}$	maximum frequency	SHCP or STCP; see Fig. 7 and Fig. 8						
		$V_{CC} = 2.0 \text{ V}$	20			16		MHz
		$V_{CC} = 4.5 \text{ V}$	24			20		MHz
		$V_{CC} = 6.0 \text{ V}$	28			24		MHz
C_{PD}	power dissipation capacitance	$f_i = 1 \text{ MHz};$ $V_I = \text{GND to } V_{CC};$ [5][6]		55				pF

[1] Typical values are measured at $T_{\text{amb}} = 25 \text{ °C}$.

[2] t_{pd} is the same as t_{PLH} and t_{PHL} .

[3] t_{en} is the same as t_{PZL} and t_{PZH} .

[4] t_{dis} is the same as t_{PLZ} and t_{PHZ} .

[5] C_{PD} is used to determine the dynamic power dissipation (P_D in μW).

$P_D = C_{PD} \times V_{CC}^2 \times f_i \times N + \Sigma(C_L \times V_{CC}^2 \times f_o)$ where:

f_i = input frequency in MHz;

f_o = output frequency in MHz;

C_L = output load capacitance in pF;

V_{CC} = supply voltage in V;

N = number of inputs switching;

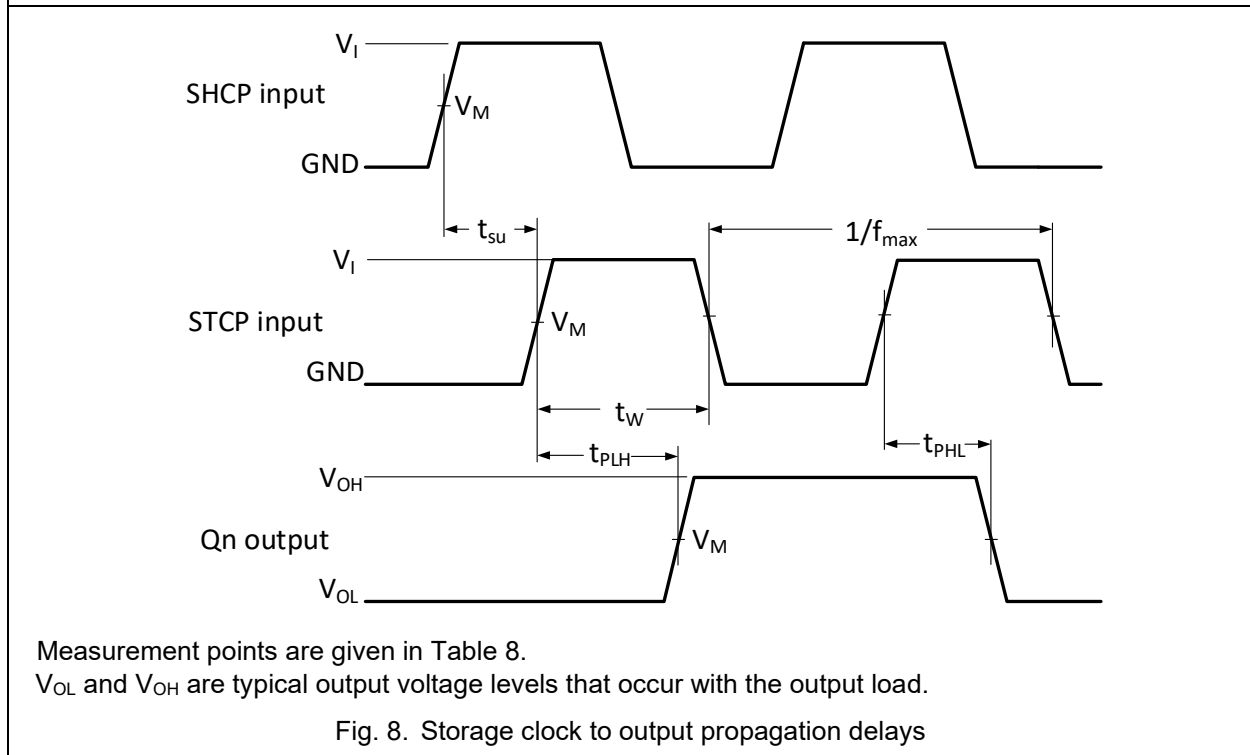
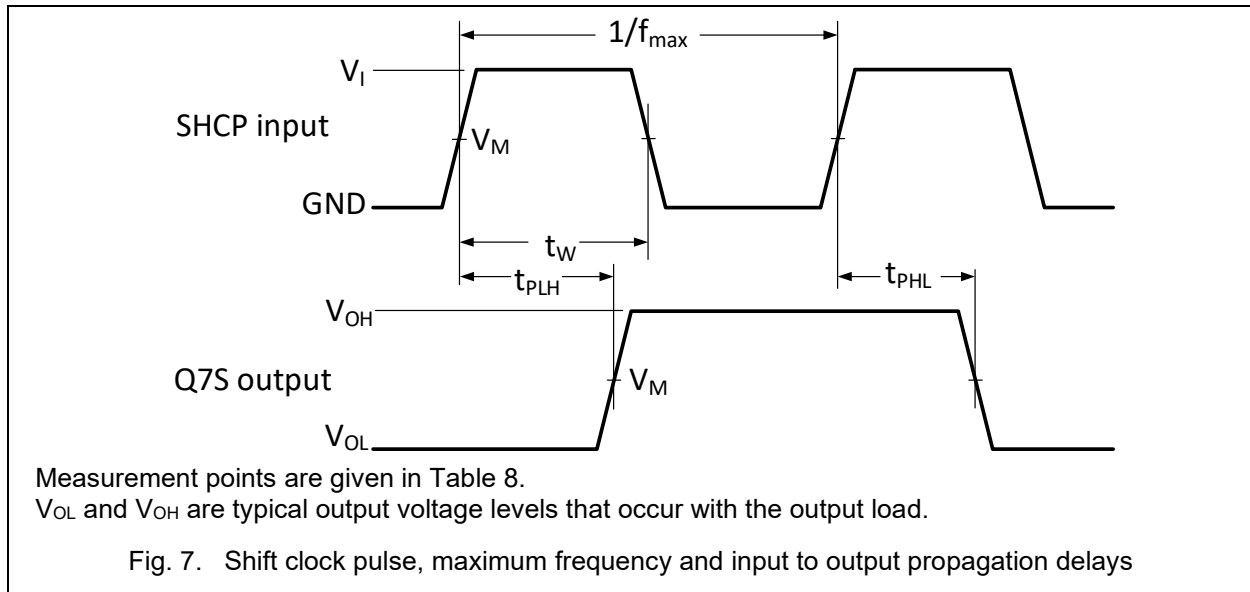
$\Sigma(C_L \times V_{CC}^2 \times f_o)$ = sum of outputs.

[6] All 9 outputs switching.

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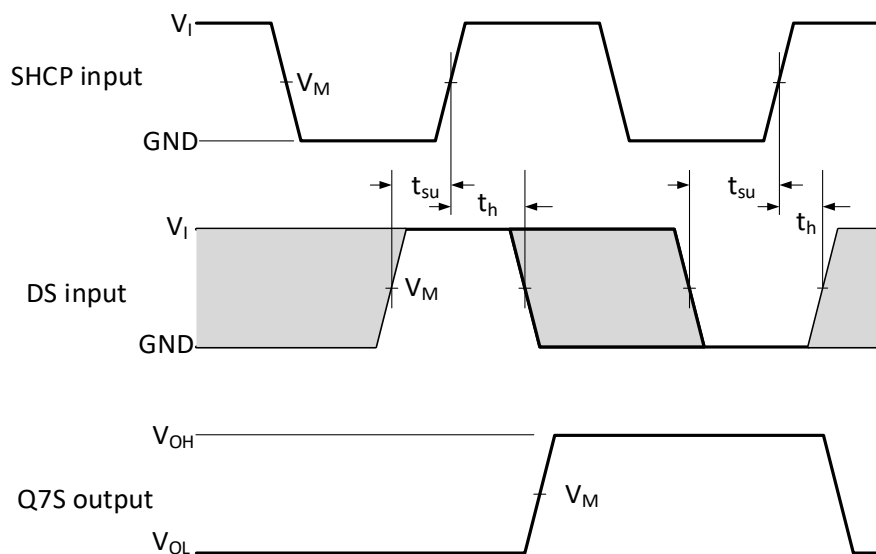
8-bit serial-in, serial or parallel-out shift register with output latches; 3-state

11.1. Waveforms and test circuit



EM74HC595A

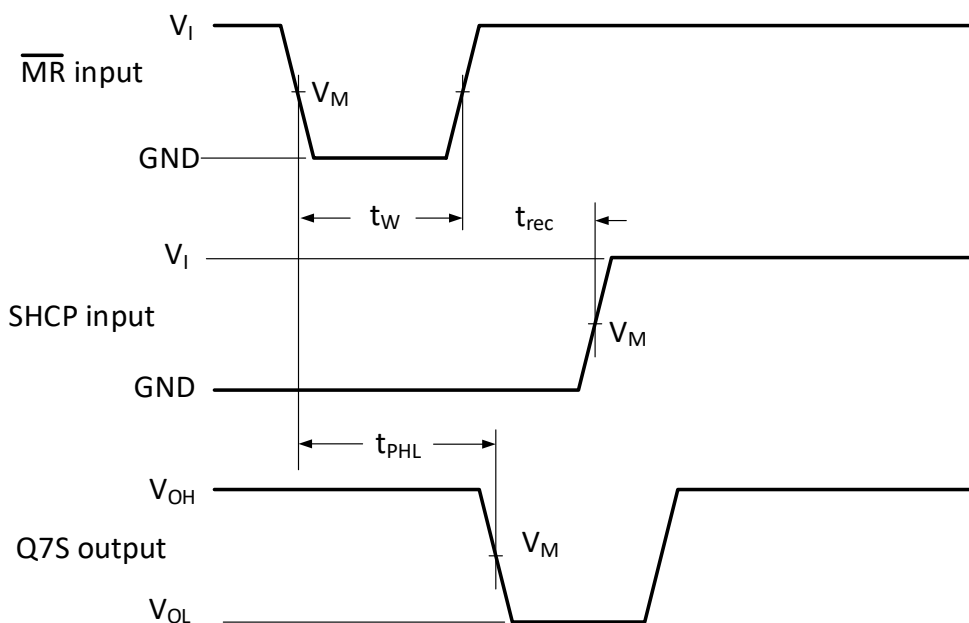
8-bit serial-in, serial or parallel-out shift register with output latches; 3-state



Measurement points are given in Table 8.

The shaded areas indicate when the input is permitted to change for predictable output performance. V_{OL} and V_{OH} are typical output voltage levels that occur with the output load.

Fig. 9. Data set up and hold times



Measurement points are given in Table 8.

V_{OL} and V_{OH} are typical output voltage levels that occur with the output load.

Fig. 10. Master reset to output propagation delays

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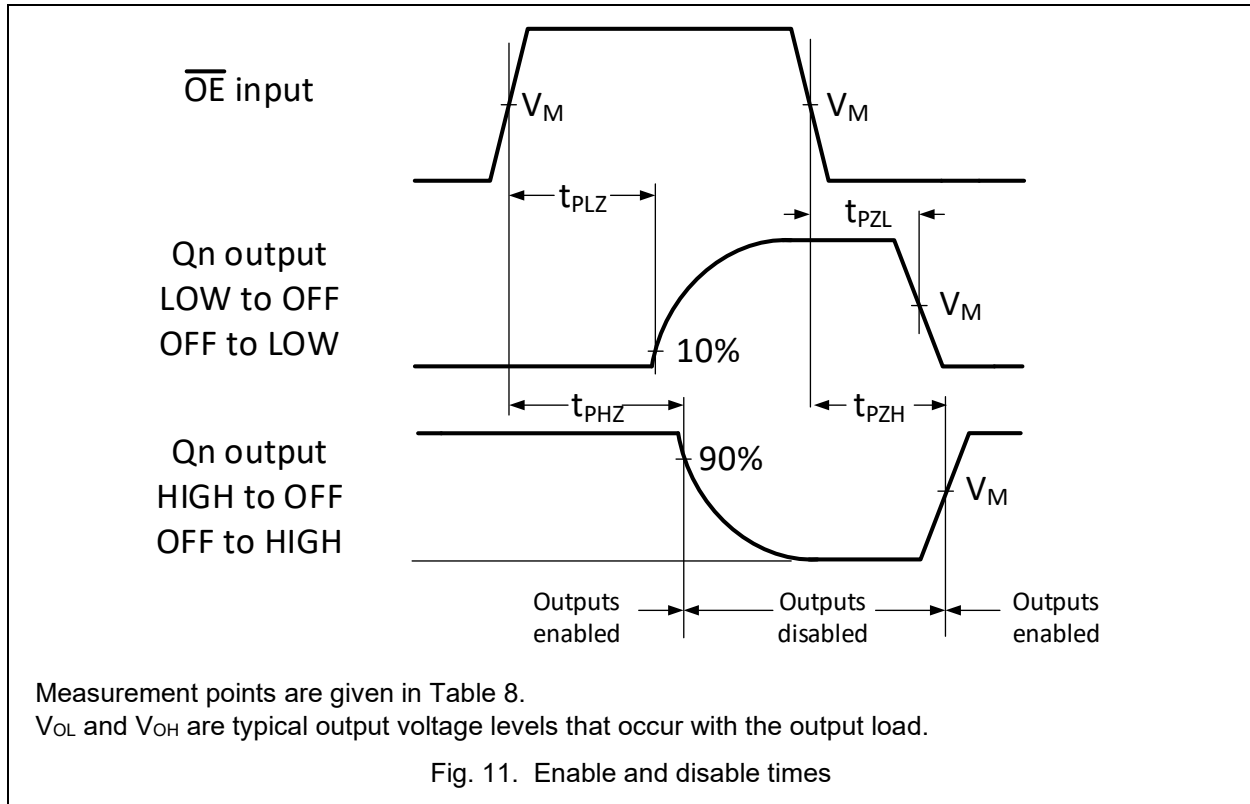


Table 8. Measurement points

Type	Input	Output
	V_M	V_M
EM74HC595A	$0.5V_{CC}$	$0.5V_{CC}$

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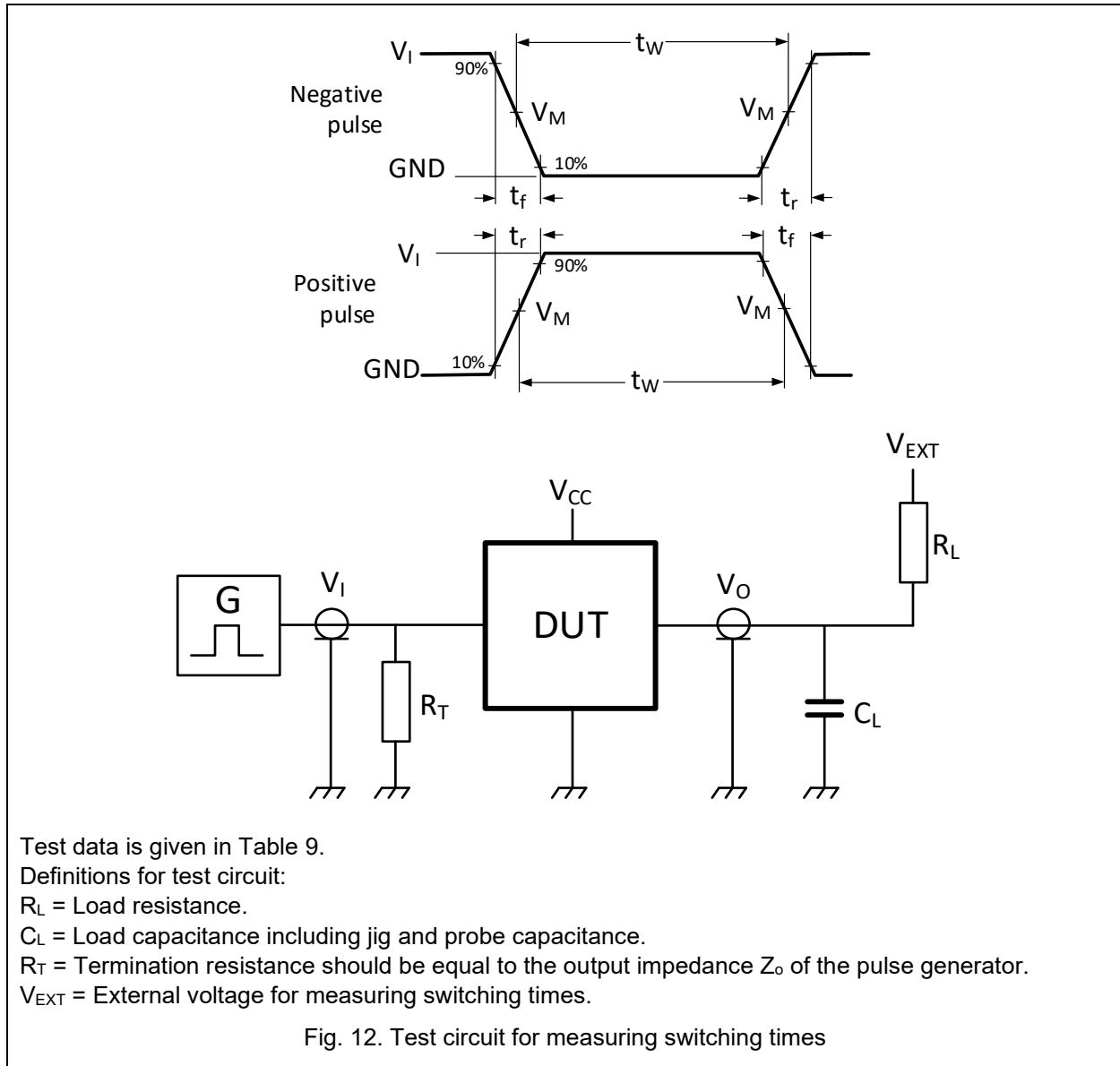


Table 9. Test data

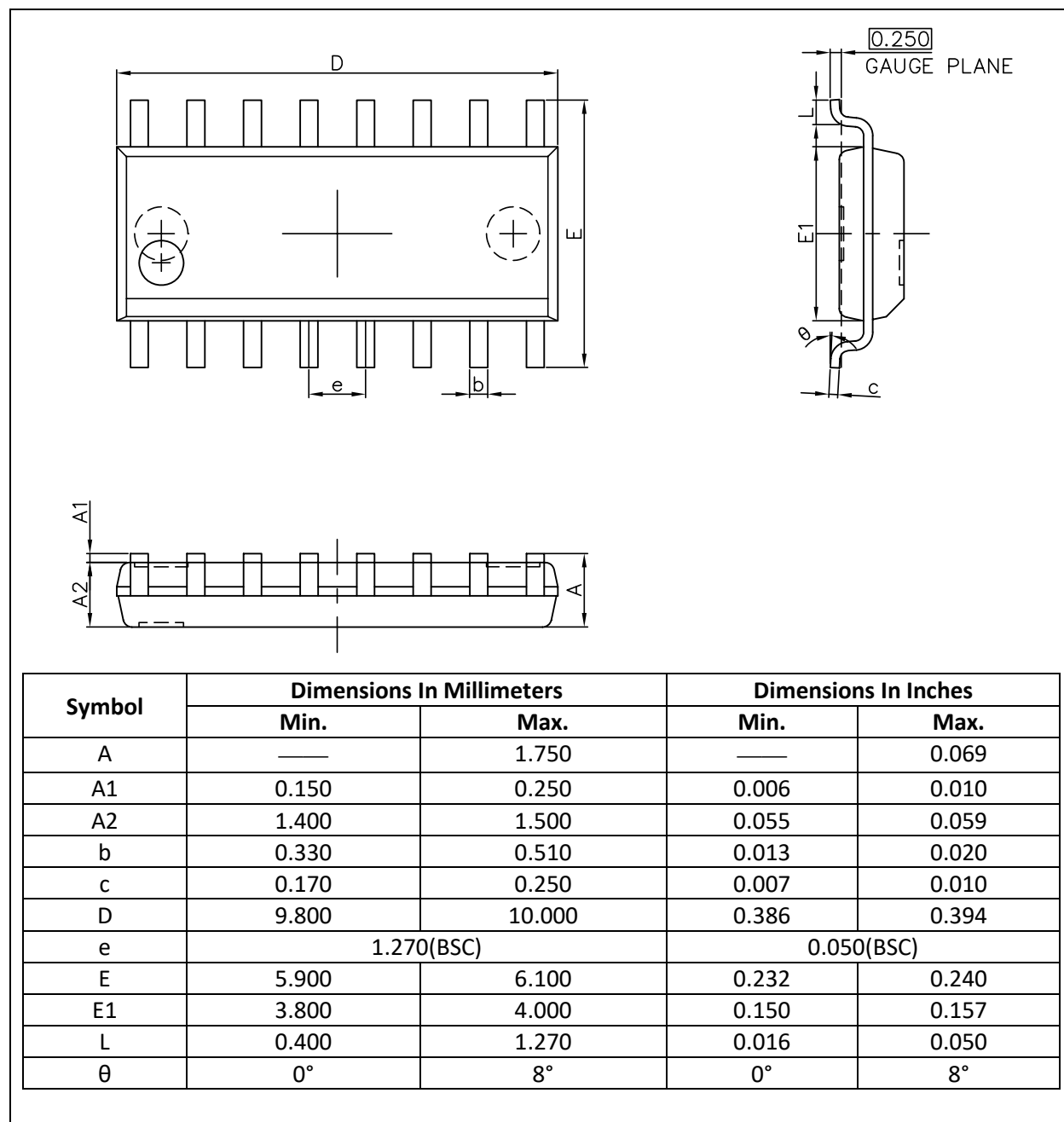
Type	Input		Load		V_{EXT}		
	V_I	$t_r = t_f$	C_L	R_L	t_{PZL}, t_{PLZ}	t_{PHL}, t_{PLH}	t_{PZH}, t_{PHZ}
EM74HC595A	V_{CC}	2.5 ns	50 pF	500Ω	V_{CC}	Open	GND

EM74HC595A

8-bit serial-in, serial or parallel-out shift register with output latches; 3-state

12. Package Outline

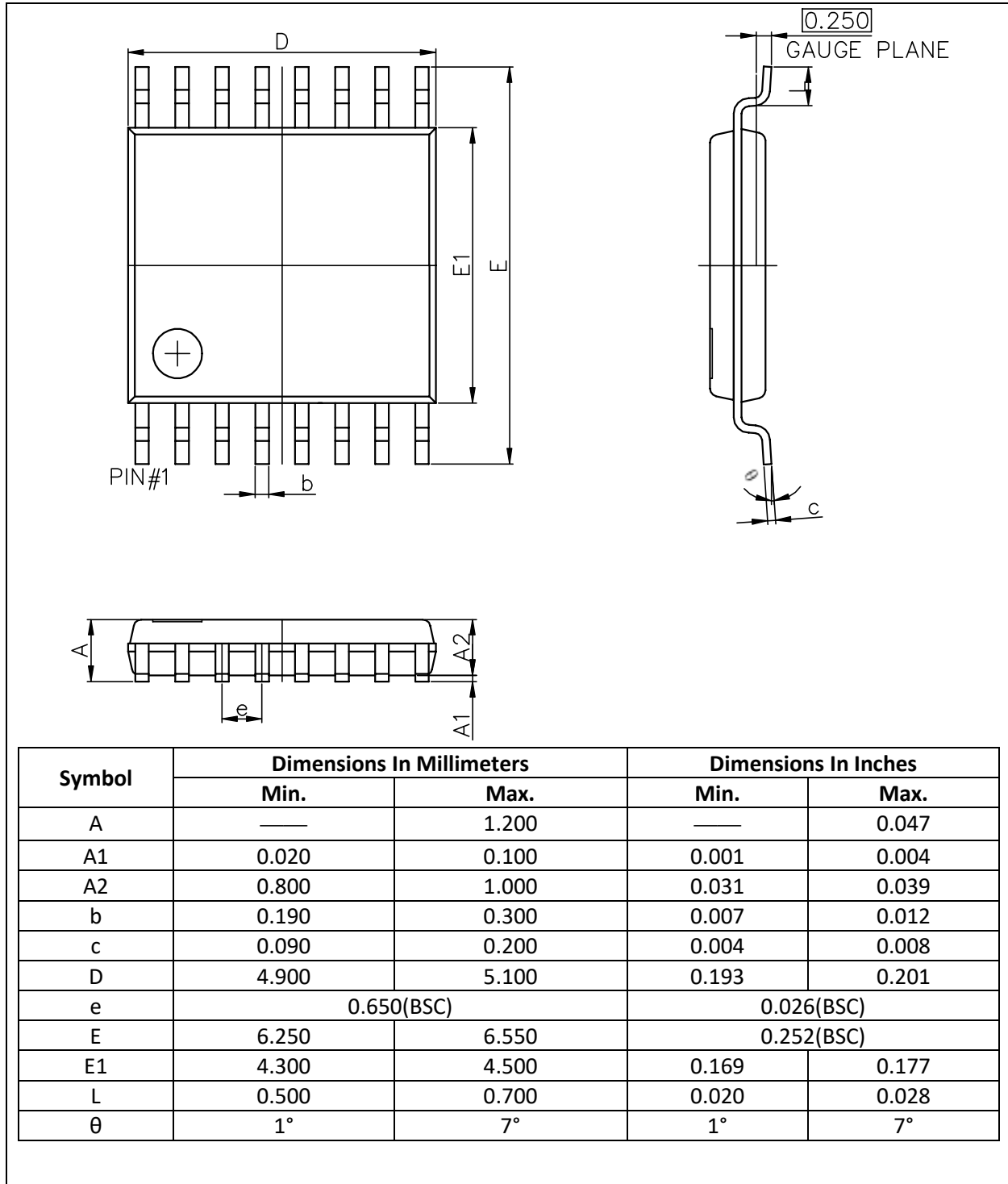
SOP-16L



EM74HC595A

8-bit serial-in, serial or parallel-out shift register with output latches; 3-state

TSSOP-16L



EM74HC595A

8-bit serial-in, serial or parallel-out shift register with output latches; 3-state

13. Tape and Reel Information

13.1. Carrier tape dimensions

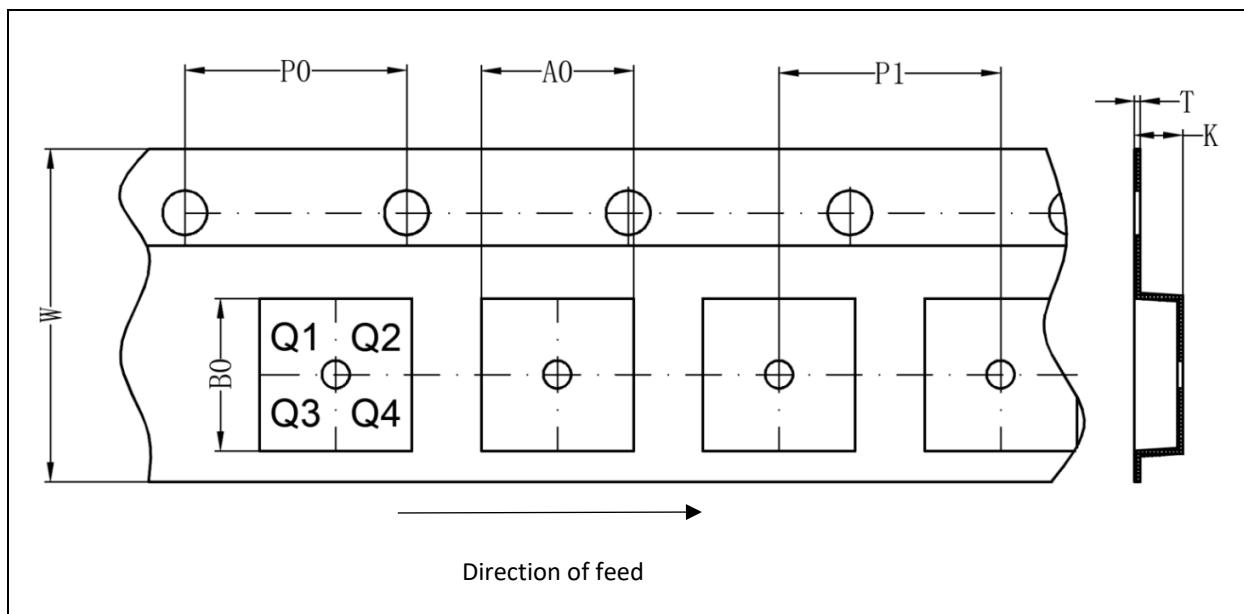


Table 10. Carrier tape dimensions

Package version	A0(mm)	B0(mm)	K0(mm)	T(mm)	P1(mm)	W(mm)	P0(mm)	PIN 1
SOP-16L	6.5	10.45	2.1	0.22	8	16	4	Q1
TSSOP-16L	6.7	5.45	1.6	0.25	8	12	4	Q1

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13.2. Reel and box dimensions

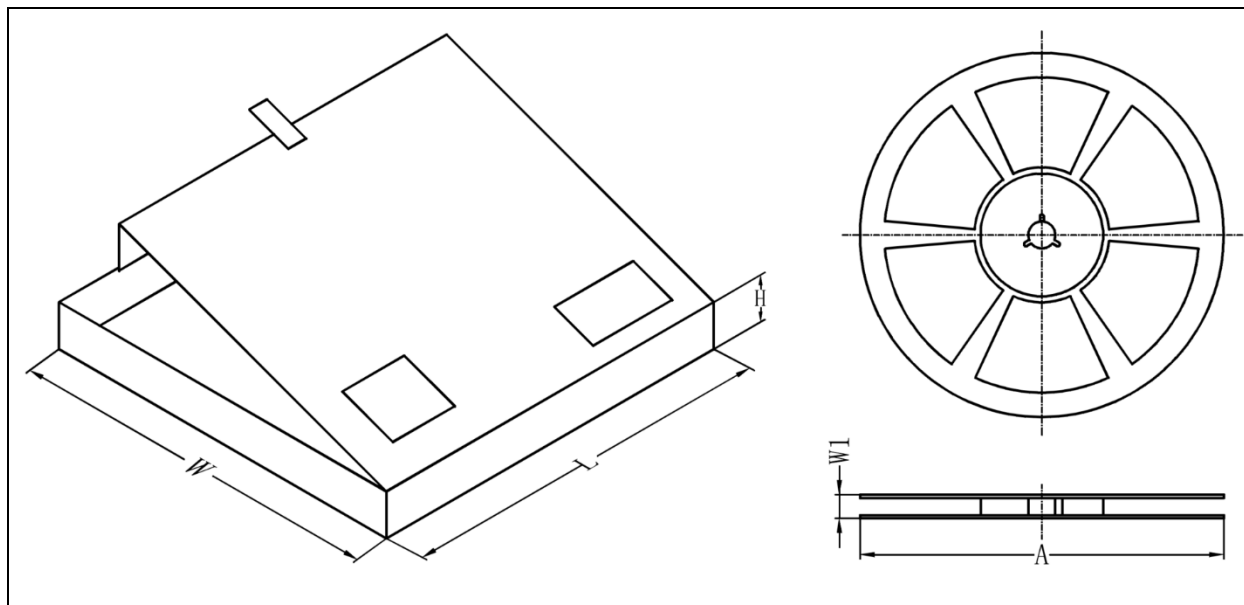


Table 11. Dimensions and quantities

Package version	Type NO. ending	Reel Dimension A (mm)	Reel Width W1 (mm)	MPQ (pcs)	Reels per box	Outer box dimensions L×W×H(mm)
SOP-16L	D	330	22.4	3000	1	358x340x50
TSSOP-16L	PW	330	18.4	3000	1	358x340x50

EM74HC595A**8-bit serial-in, serial or parallel-out shift register with output latches; 3-state**

14. Abbreviations

Table 12. Abbreviations

Acronym	Description
CMOS	Complementary Metal-Oxide Semiconductor
DUT	Device Under Test
ESD	ElectroStatic Discharge
HBM	Human Body Model
CDM	Charged Device Model

15. Revision History

Table 13. Revision history

Document ID	Release Date	Data sheet status	Change notice	Supersedes
EM74HC595A Rev. 1.1	Jun 07, 2025			EM74HC595A Rev. 1.0
Modifications:	• Section 13: added tape and reel information.			
EM74HC595A Rev. 1.0	Jul 07, 2024	Product datasheet		